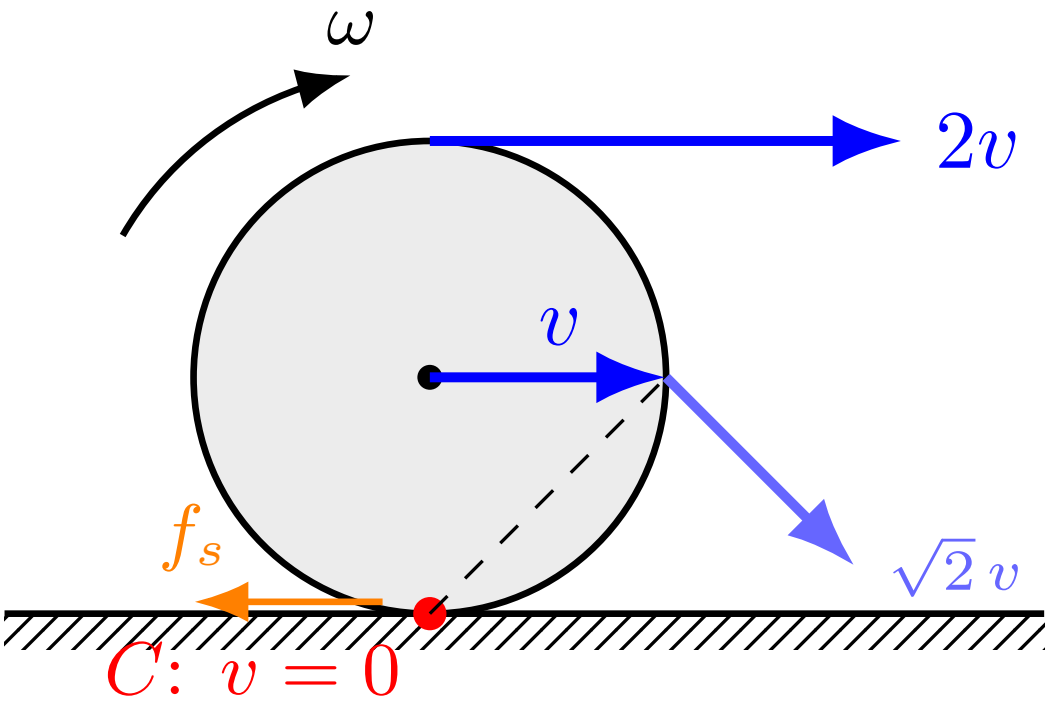

ω
$2v$
v
f_s
$\sqrt{2}\,v$
$C\colon\ v = 0$